Daniel Alejandro Silva Rojas

SENIOR BACKEND & SYSTEMS ENGINEER · PERFORMANCE · AI

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PROFESSIONAL SUMMARY

Backend & Systems Engineer with 5+ years of experience building low-latency, real-time services in Node.js, Rust, C++ and Python. At Ea2technology (Canada, until September 2026) I designed industrial Digital Twin systems and the full-duplex WebSocket architecture behind them, which took dashboard latency from 200–500 ms to 1–10 ms and cut bandwidth by up to 95%. As a consultant I audited production stacks, found the memory leaks and concurrency bottlenecks that cut a 12–14 hour pipeline to 5 hours, and built the containerized NestJS services and CI/CD pipelines around that work. I pair that with computer vision, scientific computing (NumPy/SciPy, PyTorch) and RAG systems in Python. B.Sc. in Computer Science (LUZ-IUTA) completed in parallel with continuous professional work since 2021. English B2 (professional), with five years of daily work alongside a Canada-based team.

TECHNICAL SKILLS

Languages: Rust, C++, Python, TypeScript, JavaScript (ES6+), SQL

Backend & APIs: Node.js (Express, NestJS, Nitro), FastAPI, Rust (Axum, Rocket, Tokio), REST API design, gRPC, full-duplex WebSockets, TCP sockets, custom binary/file-transfer protocols

Systems & Real-Time: Low-latency architecture, concurrency, memory safety, profiling and bottleneck analysis, industrial telemetry, Digital Twin architectures, PyO3

AI & Scientific Computing: RAG systems, AI API integration (Gemini Live), PyTorch, FlowNet / distilled deep architectures, NumPy, SciPy, OpenCV, PaddleOCR

Data: PostgreSQL (schema design, indexing, transactions, query optimization, RLS), MySQL, Redis, MongoDB, Supabase

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda), Docker, CI/CD with GitLab CI, GitHub Actions and AWS CodePipeline (CodeCommit, CodeArtifact, CodeBuild, CodeDeploy), Vercel

PROFESSIONAL EXPERIENCE

Lead Performance & Systems Engineer

Jul 2021 – Sep 2026 · full-time

Ea2technology — Canada (remote)

- Designed a low-latency, full-duplex WebSocket architecture for real-time dashboards, replacing HTTP polling with event-driven push: latency from 200–500 ms down to 1–10 ms and up to 95% less bandwidth.
- Designed and deployed an industrial Digital Twin system for real-time monitoring of coke drum deformation in a refinery, ingesting high-frequency sensor telemetry to predict and prevent critical structural failures.
- Optimized compute-intensive data pipelines in C++ and Rust, engineering for memory safety and maximum concurrency under sustained load.
- Led the migration of legacy monolithic systems to cloud-native microservices on AWS, owning service boundaries, deployment flow and the architectural trade-offs behind them.
- Led end-to-end technical UI/UX design in Figma, connecting low-level performance work to the interfaces operators actually use; worked daily in English with a distributed team based in Canada.

Software Performance Consultant & Algorithm Engineer

Jan 2023 – Jul 2025

Freelance / Independent Consulting · Remote — client: Vessel Inspection and Assessment Corp (VIAAC), maritime/oil sector · role: Senior Developer

- Built and containerized backend services with NestJS and Docker, and set up the delivery pipelines around them: GitLab CI, GitHub Actions, and AWS CodePipeline with CodeCommit, CodeArtifact, CodeBuild and CodeDeploy.
- Ran 4 technology-stack audits, identifying 3 memory leaks and concurrency bottlenecks in the processing of point clouds up to 1.2M points (3D drum meshes); the resulting optimizations cut a heavy sequential process from 12–14 hours to 5 hours and data-navigation response time from ~1 s to 15–20 ms.
- Developed high-performance video stabilization engines using optical flow and matrix interpolation (Python, NumPy, SciPy, OpenCV), smoothing trajectories via Euclidean transformations for a 65–80% jitter reduction at 60 fps from HD to Full HD.
- Implemented global motion estimation with FlowNet (distilled deep architectures), combining classical mathematics with modern machine learning; this research later became the foundation of a camera tracking system.
- Designed and deployed TCP socket and WebSocket communication systems with custom file-transfer protocols for high-frequency environments.
- Presented audit findings and remediation standards to the client's development team and specified the changes they carried into their own codebase.

Founder & Software Architect

Jun 2021 – Present · part-time personal project

DropAudio CCS — dropaudioccs.com · e-commerce platform in production

- Own the backend of a live revenue-generating store: Node.js server routes and API endpoints (Nuxt 3 / Nitro), PostgreSQL schema with Row Level Security, indexes and optimized queries, deployed on Vercel — 110+ verified customer reviews.
- Built a custom multi-currency checkout (USDT, Zinli, Pago Móvil) with live BCV/USDT exchange rates, order state handling and reconciliation, plus scheduled jobs (pg_cron) driving automated Web Push notifications.

- Engineered for operational continuity: fallback catalog for upstream outages, plus technical SEO/GEO and PWA delivery.

IT Support Specialist Jan 2021 – Sep 2021

RenéDessés de Venezuela — Caracas, Venezuela

- Sole IT/systems administrator for an 80-person company: Cisco network infrastructure, MySQL Server database, Profit business management system and Apache/Linux servers, plus day-to-day support for all 80 users.

FEATURED PROJECTS — CODE AT GITHUB.COM/DAN178A

RealtimeVoiceAssistant · **Rust (Tokio) · Python · Gemini Live**
Real-time conversational voice assistant with streaming transcription, NLP and voice synthesis on an async Rust runtime.

extract_rif · **FastAPI · PaddleOCR · Docker**
HTTP microservice that extracts structured data from RIF tax documents (SENIAT) out of PDFs or images via a dual-engine OCR pipeline.

System_Stabilization_Interpolation · **Python · PyTorch · OpenCV — degree thesis**
On-device video stabilization with deep learning and mathematical interpolation, tuned for real-time processing with explicit numerical-precision control to avoid critical rounding errors.

EDUCATION

B.Sc. in Computer Science — LUZ-IUTA (Universidad del Zulia · University Institute of Technology), Venezuela · graduated April 2025 · completed while working full-time

Associate Degree (TSU) in Information Technology — LUZ-IUTA, Venezuela · 2018 – 2021

LANGUAGES & CERTIFICATIONS

Spanish — native · **English** — B2 professional (technical reading/writing, interviews, five years of daily work with a Canada-based team)

Introduction to Generative AI — Duke University (Coursera, 2025)